INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR
PG Diploma in AI-ML and Agentic AI Engineering · Cohort 2

MINI PROJECT 1 · BUILD WEEK 1 · MODULE 1, FOUNDATIONS OF AI AND DATA

Data to Insight in Kalpa Health

A diagnostics chain grew 5 percent against a plan of 18. Find out where the growth went.

> **THE GOAL**
>
> Give the COO one sentence she can defend to her board, backed by numbers her finance team can audit.

A	B	C	D	E
GROWTH	TWO CITIES	MONEY	ONE CLINIC	THE OFFER
Which part of my business is short?	Why did bookings fall in two cities?	Why do my invoices and my collections disagree?	Is that clinic's no-show rate really worse?	Did free home collection lift bookings 9 percent?

5	3 to 4	40	1
briefs	working days	marks	public repository per group

Kalpa Group and Kalpa Health are fictional. The listed companies named in this handbook are real; every figure about them is quoted from their published results, checked on 19 October 2026.

PROJECT HANDBOOK

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How a diagnostics chain earns its money

A diagnostics chain sells tests. A patient walks into a clinic or books a home visit, a sample is taken, a lab runs one or more tests on it, and an invoice follows. Three numbers describe the whole business.

METROPOLIS HEALTHCARE, QUARTER TO JUNE 2026, AS REPORTED

Revenue	=	Patients	×	Tests per patient	×	Revenue per test
Rs 450.2 cr		3.7 million		about 2.1		Rs 575

3.7 million patients × about 2.1 tests each × Rs 575 a test comes to roughly Rs 450 crore. The tree closes.

KALPA HEALTH, FIRST HALF OF THIS YEAR

Revenue	=	Patients	×	Tests per patient	×	Revenue per test
Rs		?		?		?

Three blanks. Fill them by the end of your profiling, for both years, and the gap has a shape.

THE WORDS YOU WILL HEAR

Patient
One person who booked. Counted once per visit.

Sample
What is collected: a tube of blood, a swab. One visit can give several.

Test
One analysis run on a sample. A full-body package is many tests on one sample.

Home collection
A phlebotomist comes to the house. Costs the chain more per visit; wins patients who would not travel.

No-show
A booking where no sample was ever collected.

WHY THE THREE NUMBERS MATTER

- If patients grow but revenue does not, the chain is selling cheaper tests or fewer of them per visit.
- If revenue per test rises with no price increase, the mix shifted toward dearer tests. Chains call this realisation.
- A plan set on test volume can be missed because fewer people came, or because the same people ordered less. Those need different fixes.

What the listed chains reported for the quarter to June 2026

India's two largest listed diagnostics chains publish these numbers every quarter. A COO at any other chain is measured against them whether she likes it or not.

Dr Lal PathLabs

Rs 797.7 crore +19.1%

PATIENTS
8.2 million
+8.2%

SAMPLES
25.9 million
+10.7%

Revenue grew more than twice as fast as patients. Each patient brought more samples, and each sample earned more.

Metropolis Healthcare

Rs 450.2 crore +16.6%

PATIENTS
3.7 million
+10%

TESTS
7.8 million
+11%

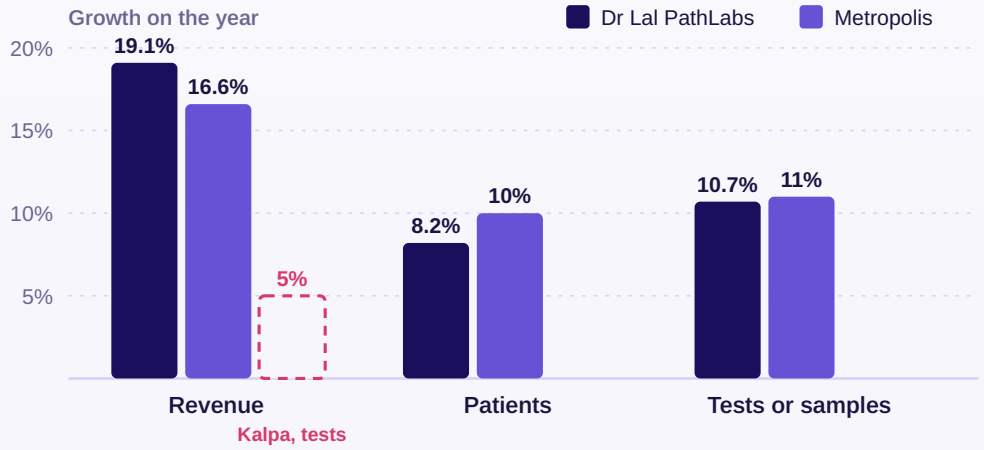
Rs 1,219 per patient and Rs 575 per test. Management said no prices were raised, so the mix of tests moved.

Kalpa Health

Not public

TEST VOLUME
Test volume +5% against a plan of +18%

A board that reads the two cards on the left will not accept that the market was soft.



WARM-UP, BEFORE THE DATA ARRIVES

1. Work out revenue per patient and per sample for Dr Lal from the card. Write down what you assumed.
2. Dr Lal counts samples and Metropolis counts tests. Say in one line why Rs 575 per test is not comparable with your per-sample figure.
3. Both chains grew money faster than people. Which branch of the tree moved, and what in Kalpa's data would show the same thing?

Sources: Metropolis Healthcare Q1 FY27 earnings call transcript filed with NSE, and its Q1 FY27 investor presentation as summarised by Investing.com; Dr Lal PathLabs Q1 FY27 results as reported by Yahoo Finance and Business Upturn. Thyrocare and Apollo Diagnostics are named as peers elsewhere in this handbook; no figure about them is used.

02 · THE CLIENT

Kalpa Health

A mid-sized chain inside Kalpa Group: diagnostic laboratories and walk-in clinics in six Indian cities, two channels, one COO who answers to a group board in Singapore.

6 cities

2 channels: clinic and home collection

1 central lab per city

5% test volume growth, first half

18% the plan

Competes for the same patients as Dr Lal PathLabs, Metropolis Healthcare, Thyrocare and Apollo Diagnostics.

WHERE THE DATA COMES FROM

Booking system

Takes bookings from the app, the phone line and the front desk. Two cities moved to a new vendor in the second quarter.

`bookings`

Collection app

Phlebotomists mark a sample collected at the clinic or at the door, or mark the patient absent.

`collections`

Lab system

Runs each test, records the result, timestamps it.

`results`

Billing

Raises one invoice per visit from the tests performed.

`invoices`

Payment gateway

Settles card, UPI and cash deposits. Its ids look nothing like billing's.

`payments`

Marketing tool

Tags which bookings came under an offer.

`campaigns`

The data pack The data pack is an export of these six systems plus the patient master and the city plan. Nothing in it has been cleaned.

> *A count is only as true as the system that recorded it.*

02 · THE CLIENT

The situation: a board meeting in two weeks

The group board meets in Singapore in two weeks and Kalpa Health is on the agenda. The dashboard shows one number a month. Dr Menon has to turn that number into an explanation.

LAST YEAR
The plan is set: 18 percent growth in test volume, city by city, on last year's mix.

FIRST HALF
Test volume lands at 5 percent. Two cities fall. A free home collection offer runs. Billing and collections drift apart.

NOW
Five explanations are on the table. None has a number behind it.

IN TWO WEEKS
The board asks one question: where did the 13 points go, and what are you doing about it?

THE FIVE PAIN POINTS, AND THE BRIEF THAT OWNS EACH

A
Nobody can say which part of the business is short.
BRIEF A · GROWTH

B
Two city heads say demand is soft. Nobody has checked.
BRIEF B · TWO CITIES

C
Billing and collections disagree, and have for a while.
BRIEF C · MONEY

D
One clinic looks terrible on no-shows. Its manager says bad luck.
BRIEF D · ONE CLINIC

E
Marketing wants its offer budget doubled on a 9 percent lift it measured itself.
BRIEF E · THE OFFER

WHO IS IN THE ROOM

Dr Priya Menon COO
Needs one defensible sentence per pain point.

The finance controller FINANCE
Will audit every number back to rows, and reject what cannot be traced.

The city heads OPERATIONS
Will push back if blamed, harder if their people are.

The marketing lead GROWTH
Will defend the 9 percent.

You ANALYST
Say what you found, how sure you are, what to do.

Objective, targets and scope

OBJECTIVE

Answer one of the five pain points end to end: a claim Dr Menon can repeat to the board, the evidence behind it with every denominator shown, the one caveat that would change it, and the action with its cost.

TARGETS EVERY GROUP IS HELD TO

T1 **Reproduce the 5 percent**
Your test-volume growth matches Dr Menon's figure, or you can say exactly why it does not.

T2 **One sentence**
A headline claim with a number, a denominator and a period, that a COO can say aloud.

T3 **Auditable**
Every number traces to rows; every cleaning decision is logged with its effect on the headline.

T4 **Reproducible**
The notebooks run top to bottom from a fresh Codespace in under ten minutes.

T5 **Defended**
Every member can explain every number and every approach the group rejected.

IN SCOPE

- The data pack as released, first half of this year against the same half of last year.
- Your brief, answered end to end.
- Python notebooks, SQL for joins, Excel for the one board number.

OUT OF SCOPE

- Forecasting next quarter.
- Fixing Kalpa Health's systems.
- Data you do not have. Say so in the caveat.
- Dashboards and web apps, unless everything else is done.

DELIVERABLES

Six slides
question, picture, claim, evidence, caveat, action

Notebooks
numbered, outputs cleared, run cold

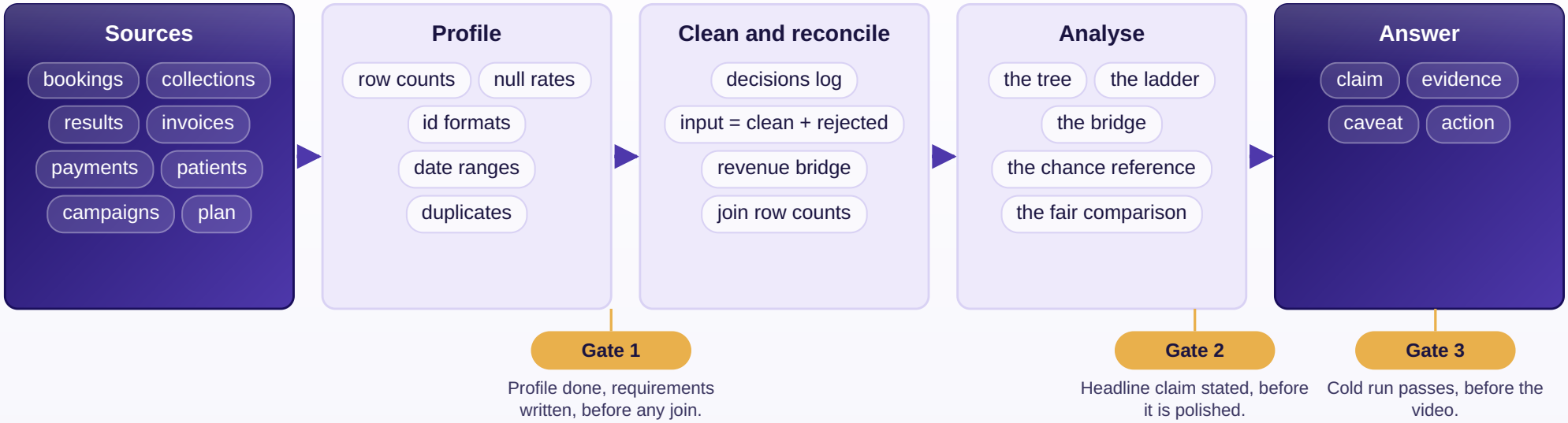
Three logs
approach, decisions, challenges

Write-ups
one section per member

Video
each member on their own part

The analysis, as a system

Every brief runs the same pipeline. What changes is which tables enter and which question comes out.



RUNS BESIDE EVERY LAYER

Approach log
two options per part, the one chosen, why

Challenges log
what broke, what you tried, what changed

Assistant log
every use, one line

Requirements, in plain words

What the analysis must do, how well, and what it may not use. Write your own rows before the data arrives.

Functional

What it must do

- Reproduce the 5 percent or explain the gap.
- Answer the brief with a number and its denominator.
- Produce the list or split someone can act on.

YOUR ROWS

Non-functional

How well

- Cold run under ten minutes, no manual edits.
- A finance analyst can audit every number to rows.
- Every cleaning decision logged before the next step.

YOUR ROWS

Constraints

What you cannot change

- Free tools covered so far only.
- No assistants on analysis, notebooks or write-ups.
- The data as released; three to four days; a group of three or four.

YOUR ROWS

> *Write requirements from Dr Menon's side of the table. She does not require a notebook. She requires a sentence she can defend.*

Three to four working days, in four phases

Plan backwards from the presentation. The percentages are of your working hours, whichever days you have.



Share of your working hours, with a checkpoint at each gold dot

1 Scope

Translate the brief, write requirements, predict the answer, open the repository and the logs.

OUT

Translation note, requirements table, prediction, approach log row 1.

2 Profile and clean

Profile every table before touching it. Decide, log, reconcile. Join with row counts.

OUT

Decisions log, reconciliation checks passing, first honest totals.

3 Analyse

Run the brief's method. State the claim early and spend the rest testing it.

OUT

Headline claim with denominator and caveat; approach log complete.

4 Answer and defend

Six slides, cold run, rehearsal, write-ups, video.

OUT

Repository frozen; video linked; every member can present any slide.

CHECKPOINT QUESTIONS AT EVERY PHASE BOUNDARY (THE GOLD DOTS)

1. What is the headline number right now, with its denominator?
2. What did we reject since the last checkpoint, and why?
3. What would make us wrong, and what are we doing about it?

> *Groups that skip Scope rewrite their analysis in Answer. Groups that skip Profile present a number they cannot defend.*

First principles, then five whys

STRIP THE QUESTION TO WHAT MUST BE TRUE

1. Test volume is a count of tests. A test needs a booking, a collected sample and a result. Which does your table record?
2. A growth rate is two numbers and a period. Name all three or you do not have one.
3. Nothing is true because the header says so. It is true when two tables agree.

ASK YOURSELF BEFORE YOU ASK ANYONE

1 What am I taking as true without checking?	2 What is the denominator?
3 What would I see if I were wrong?	4 Who loses if I am right?
5 Which number am I surest of, and why?	6 What is the cheapest way to find out?
7 Why would another group get a different answer?	8 What did Dr Menon not ask?

FIVE WHYS ON DR MENON'S PROBLEM

ILLUSTRATION OF THE METHOD, NOT THE ANSWER

1. **Why is Kalpa Health 13 points short?**
   Test volume grew 5 percent against 18.
2. **Why did volume grow so little?**
   Fewer patients, fewer visits, or fewer tests per visit. Nobody has split it.
3. **Why has nobody split it?**
   The dashboard reports one number; the plan was set per city on last year's mix.
4. **Why one number?**
   The board asked for one, and nobody asked what they would do with it.
5. **Why did nobody ask?**
   The question only hurts when the number is bad.

Five whys finds the question the data has to answer. Write your group's chain before you open the data.

The approach log: why this, and not that

For every part of the work: two ways you could have done it, the one you chose, and why. The panel reads this before your notebook.

PART	APPROACHES CONSIDERED	CHOSEN	WHY THIS AND NOT THE OTHERS	WHAT WOULD CHANGE THE CHOICE	WHAT HAPPENED
Invoice id to payment reference (brief C)	(a) strip prefix, match digits. (b) match amount and date within two days. (c) both; flag disagreements.	(c)	(a) alone paired two invoices with one payment. (b) alone could not separate same-day same-amount visits.	A shared id in both feeds.	96 percent matched; the rest became finance's list.

HOW TO KEEP IT

- Lives at docs/approach-log.md; every member commits to it.
- Written when the choice is made. The commit time is the proof.
- A rejected approach you tried for twenty minutes is the best entry. Keep the dead cell.
- One row per part: profiling, cleaning, each join, each comparison, each chart.

WHAT THE PANEL DOES WITH IT

A panellist covers the chosen column and asks you to choose again, aloud. If you now pick differently, the panel asks which of you was wrong.

Which part of my business is short?

THE SITUATION

Last year Kalpa Health's board approved a plan for 18 percent more tests in the first half of this year. The plan was set city by city on last year's mix of patients and tests, and the budget for staff, reagents and machines was built on it.

The half closed at 5 percent. The dashboard shows one monthly total, so nobody can say whether fewer people came, the same people came less often, or each visit carried fewer tests. Finance says the shortfall is already costing margin because the fixed costs were sized for 18.

Dr Lal reported 19 percent growth and Metropolis 17 for the overlapping quarter. The board will not accept a soft market.

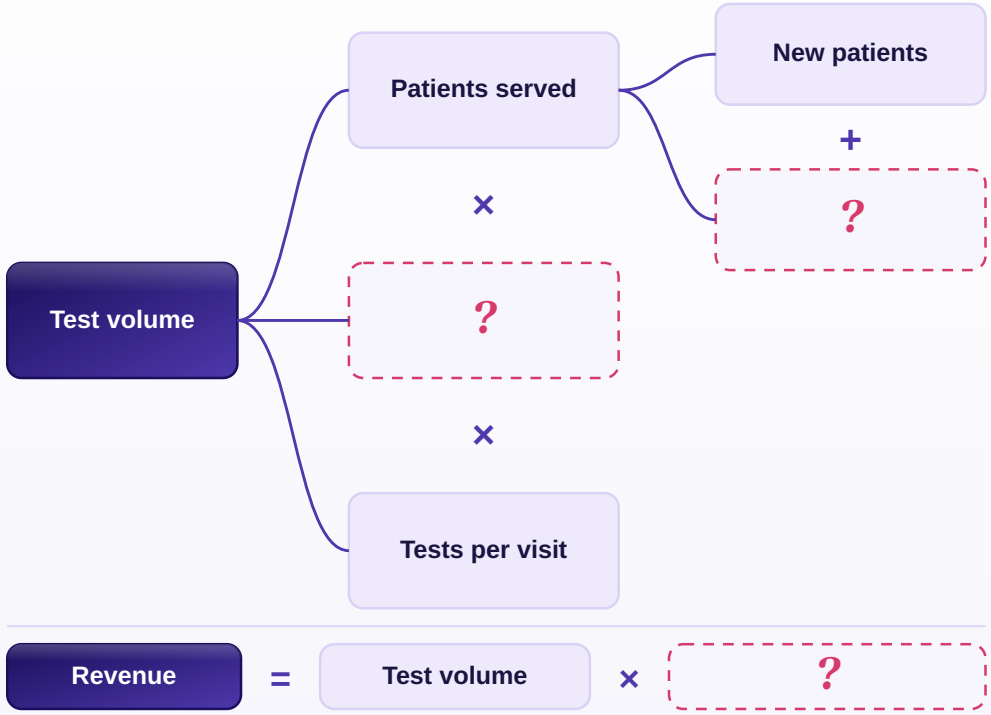
> *"Show me the part of my business that is short, by how much, and where."*
>
> DR PRIYA MENON, COO, KALPA HEALTH

PAIN POINTS

- One number, no branches
- Plan set on last year's mix
- Costs sized for 18

TARGET

The 13-point gap split across the branches of the tree and across cities and channels, multiplying back to the total, with one sentence for the board.



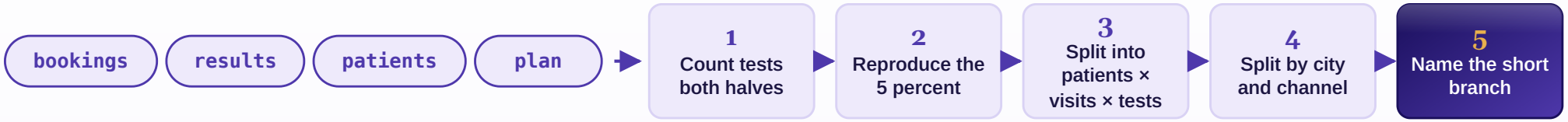
Three boxes are blank. Fill them, then put a number for both years in every box.

IN THE INDUSTRY

Metropolis earned about 5 percent more per test in the quarter to June 2026 with no price rise, and Dr Lal grew revenue 19 percent on 8 percent more patients. If Kalpa's revenue per test moved, what in your data separates a price change from a change in which tests were ordered?

Metropolis Q1 FY27 call, filed with NSE; Dr Lal PathLabs Q1 FY27 results. Checked 19 October 2026.

The pipeline, then four steps with less help each



1 STEP 1 · FULLY GUIDED

Rebuild test volume for both halves and check your growth against the 5 percent. If it differs, find out why before going on, and log it.

Done when you hold two totals, one rate, and a note on any difference.

> WHY THIS, AND NOT THAT? Which table did you count from, and why not the other two that could give a count?

2 STEP 2 · WITH A HINT

Split the growth into the branches of your tree.

Hint: multiply branch growth rates back together. If they miss the total, a branch is missing or a denominator is wrong.

> WHY THIS, AND NOT THAT? Why patients × visits × tests rather than patients × tests? What does the middle branch reveal?

3 STEP 3 · THE QUESTION ONLY

The plan was set city by city. Which cities, and which channel, own most of the gap?

> WHY THIS, AND NOT THAT? Why split by city before channel? Would the order change the answer?

4 STEP 4 · OPEN

What would make you wrong? If the gap sits in one or two cities, how do you know it is a business problem and not a recording problem?

> WHY THIS, AND NOT THAT? Which single check would you run with ten minutes left?

DONE LOOKS LIKE

Every branch numbered for both years; the branch and cities that own most of the gap with their share; an approach-log row per join and split; one sentence for the board.

PUSH FURTHER

If the short branch recovered to plan, how many more tests would run, and on what assumption?

THE VIVA WILL ASK

1. Why these branches for a lab, and which retail branch did you drop?
2. Your branches multiply back to the total. Show me, now.
3. What changes when an error is a missed diagnosis rather than a missed sale?

Why did bookings fall in two cities?

THE SITUATION

Two of Kalpa Health's six cities show bookings falling in the second quarter while the other four held. The city heads say demand softened: Thyrocare opened collection centres in both cities this year and prices there are lower.

During the same quarter both cities moved from the old booking software to a new vendor. The migration was rushed; nobody wrote down how the new system counts a visit with two sample tubes, or a rescheduled appointment.

Dr Menon is deciding whether to cut the two cities' targets and headcount. If the fall is real, that is the right call. If it is a counting artefact, she would be cutting a business that is fine.

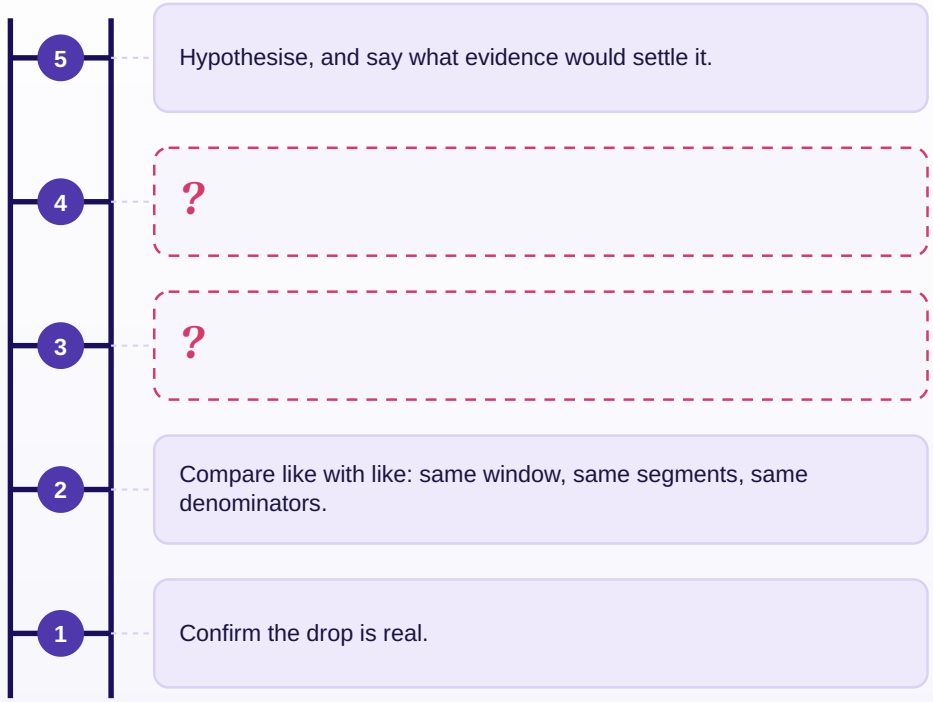
> *"Before I cut targets or people, I need to know whether the fall is real."*
>
> DR PRIYA MENON, COO, KALPA HEALTH

PAIN POINTS

- Two systems, one metric
- A plausible competitor story
- A headcount decision waiting

TARGET

The fall in each city split into a counting part and a real part, in bookings and in percent, with the real part traced to a branch of the tree.



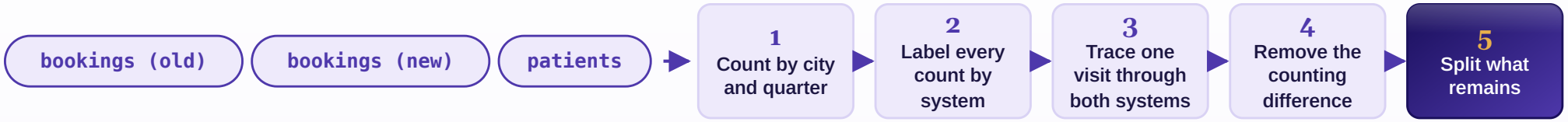
Two rungs are missing from the ladder you climbed in Week 1. Fill them before you start.

IN THE INDUSTRY

When Google replaced Universal Analytics with GA4, its help pages warned that session counts would differ, because the old product restarted a session at midnight and on every campaign change and the new one does not. A site that switched saw its sessions move with no change in visitors. What is Kalpa Health's midnight rule?

Google Analytics Help, comparing metrics between Universal Analytics and GA4. Checked 19 October 2026.

The pipeline, then four steps with less help each



1 STEP 1 · FULLY GUIDED

Count bookings by city and quarter, labelling every count with its source system.

Done when every number in your table names its system.

WHY THIS, AND NOT THAT? Why bookings, and not patients or tests? What would each of those have hidden?

2 STEP 2 · WITH A HINT

Check whether the two systems count a booking the same way.

Hint: write out how one patient's visit appears as rows in each. Different shapes, different counts.

WHY THIS, AND NOT THAT? Why trace one visit by hand rather than compare totals?

3 STEP 3 · THE QUESTION ONLY

With the counting difference removed, how much of the fall remains, and does it sit in patients, visits per patient or tests per visit?

WHY THIS, AND NOT THAT? Why remove the counting difference before splitting?

4 STEP 4 · OPEN

What would make you wrong? Name two explanations outside the data you cannot rule out, and what you would ask the city heads.

WHY THIS, AND NOT THAT? Which one question to a city head settles the most?

DONE LOOKS LIKE

Each city's fall split into counting and real, the real part on a branch, the system mapping and its rejected alternatives in the approach log, two sentences the city heads would accept.

PUSH FURTHER

Write the rule Kalpa Health should follow the next time a city changes systems.

THE VIVA WILL ASK

1. Two systems export the same booking with different identifiers. How did you reconcile them, and what did you throw away?
2. What share of the fall was counting, and how did you prove it?
3. Sales dropped 15 percent last month. Walk me through it.

Why do my invoices and my collections disagree?

THE SITUATION

Kalpa Health's billing system raises one invoice per visit. Money arrives through a payment gateway for cards and UPI and through cash deposits at the clinics. The gateway's ids look nothing like billing's, and nobody has reconciled the two since the gateway was switched.

For the second quarter, billing shows one figure invoiced and the payment feed shows a smaller figure collected. The finance controller cannot tell whether the difference is invoices never paid, payments that match no invoice, duplicate settlement rows, or simply money that arrived after the quarter closed.

Until she can, every revenue figure Kalpa Health reports upward carries an asterisk, and the board hates asterisks.

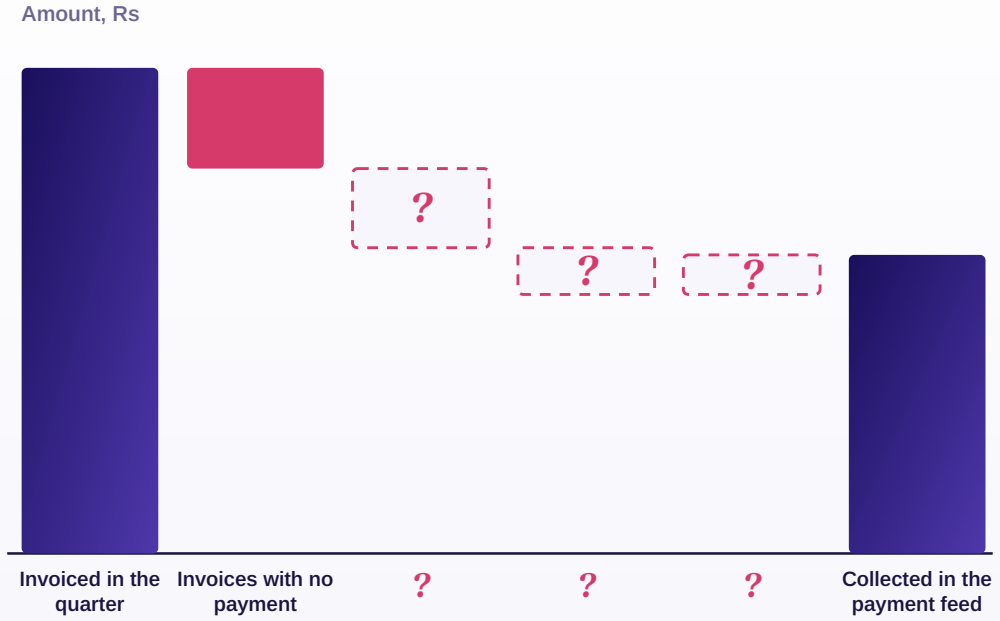
> *"Which invoices were never paid, which payments match no invoice, and how much of the gap is only timing?"*
>
> DR PRIYA MENON, COO, KALPA HEALTH

PAIN POINTS

- Two id families
- A gap nobody has sized
- Revenue reported with an asterisk

TARGET

A bridge from invoiced to collected in which every bar is a count and an amount, both sides reconcile, and finance has a list to act on.



One bar is named for you. Name the others from your data; every bar is a count and an amount.

IN THE INDUSTRY

Razorpay's documentation gives orders, payments and settlements separate id families (order_, pay_, setl_), and its sample reconciliation entry shows a Rs 1,000 payment carrying a Rs 29 fee. One settlement bundles many payments. Which of these explains a gap that is not a leak?

Razorpay API documentation, settlements and settlement reconciliation. Checked 19 October 2026.

The pipeline, then four steps with less help each



1 STEP 1 · FULLY GUIDED

Profile both exports before joining: counts, totals, id formats, date ranges, duplicates.

Done when you hold a profile of each and a written mapping rule.

> WHY THIS, AND NOT THAT? Why profile first? What would a join have done to the duplicates?

2 STEP 2 · WITH A HINT

Join invoices to payments and explain the row count before and after.

Hint: more rows than invoices means something on the payment side repeats.

> WHY THIS, AND NOT THAT? Why LEFT and not INNER? Which rows would INNER have made vanish?

3 STEP 3 · THE QUESTION ONLY

Which invoices were never paid, which payments match no invoice, and how much is timing across the quarter boundary?

> WHY THIS, AND NOT THAT? Why is timing its own bar? What happens to your leak figure without it?

4 STEP 4 · OPEN

What would make you wrong? Your bridge balances. Name one way it could balance and still be wrong.

> WHY THIS, AND NOT THAT? Which bar do you re-check first if finance says the total is off by Rs 2 lakh?

DONE LOOKS LIKE

A bridge whose bars are counts and amounts, input equals matched plus unmatched on both sides, the mapping rule and its alternatives logged, and a list finance can act on tomorrow.

PUSH FURTHER

Turn the bridge into a weekly check. Which numbers compare, and what triggers an alert?

THE VIVA WILL ASK

1. Walk me through the join. What did the row count do?
2. How do you know your unmatched list is complete?
3. The dashboard and Finance disagree. What do you do first?

06 · BRIEF D · ONE CLINIC

Is that clinic’s no-show rate really worse?

THE SITUATION

A no-show is a booking where no sample was ever collected. Every no-show is a slot a phlebotomist waited in, a kit opened, and a patient who may now be at Apollo across the road.

One clinic’s no-show rate looks far worse than the other clinics’. Its manager says it is bad luck and a small clinic has small numbers. The regional head wants the manager replaced.

Dr Menon has seen clinics ranked before and watched the ranking flip the next quarter. She wants to know whether this difference is real, noise, or not yet knowable, before anyone’s job depends on it.

> *“Tell me whether the difference is real, and if it is, whom it affects.”*
>
> DR PRIYA MENON, COO, KALPA HEALTH

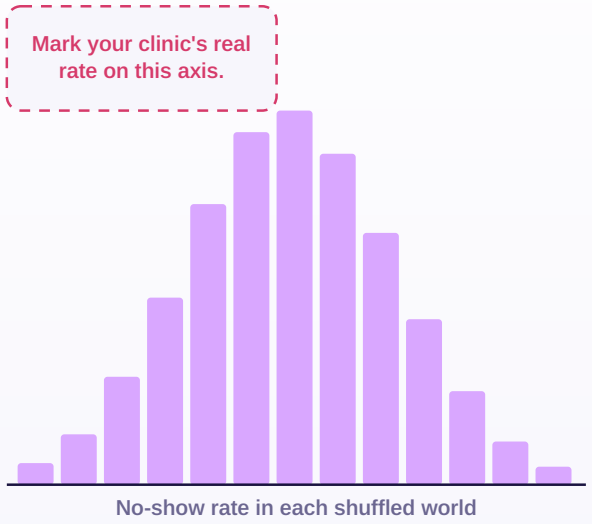
PAIN POINTS

- A ranking with no denominators
- A small clinic at the bottom
- A manager’s job on the line

TARGET

A verdict of real, noise, or not yet knowable, with the definition, the denominators, a like-with-like comparison and a chance reference.

The chance reference



Same rate, different size

Clinic P
8 no-shows in 32 bookings
25%

Clinic Q
250 no-shows in 1,000 bookings
25%

Which 25 percent would you trust, and why?

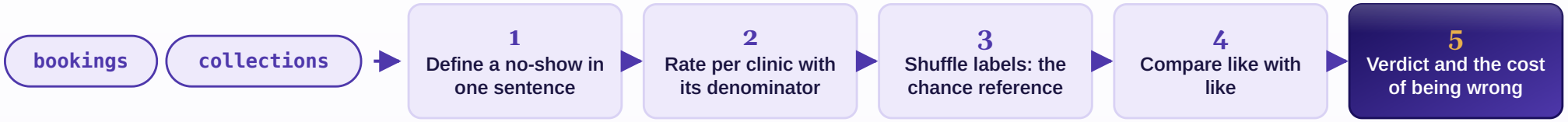
The two clinics on the right are an illustration, not Kalpa Health data.

IN THE INDUSTRY

In 2007 Tu and Gilthorpe showed that English hospital league tables put small trusts at both the top and the bottom, because small numbers swing more by chance, and recommended plotting rate against size. Wainer found the same pattern behind the Gates Foundation’s small-schools push. How large is your clinic?

Tu and Gilthorpe, BMC Health Services Research 7:185 (2007); Wainer, Picturing the Uncertain World (2009). Checked 19 October 2026.

The pipeline, then four steps with less help each



1 STEP 1 · FULLY GUIDED

Define a no-show in one sentence and apply it to every clinic. Report each rate with numerator and denominator beside it.

Done when no rate appears without its count of bookings.

> WHY THIS, AND NOT THAT? Why that definition? Name a booking it counts that someone else would not.

2 STEP 2 · WITH A HINT

Ask how often a clinic this size would look this bad by chance alone.

Hint: shuffle clinic labels across bookings many times and see where the real rate falls.

> WHY THIS, AND NOT THAT? Why shuffle rather than quote a formula? What does the shuffle assume?

3 STEP 3 · THE QUESTION ONLY

Is the clinic still worse like with like: same slots, same tests, same channel?

> WHY THIS, AND NOT THAT? Which of the three would you drop if you could control for two?

4 STEP 4 · OPEN

What would make you wrong? Cost of flagging a clinic wrongly against cost of staying quiet when it is real.

> WHY THIS, AND NOT THAT? Which cost did you weigh more, and would Dr Menon agree?

DONE LOOKS LIKE

A verdict backed by definition, denominators, like-with-like and the chance reference; the shuffle choice logged; if not yet, the number of bookings that would settle it.

PUSH FURTHER

Plot every clinic's rate against its bookings with the band chance alone would produce. Who sits outside it?

THE VIVA WILL ASK

1. How do you know a difference is significant, and what does your p-value not mean?
2. How many bookings would settle it?
3. What does a no-show cost a patient, against a cancelled order in retail?

Did free home collection lift bookings 9 percent?

THE SITUATION

Home collection is the growth channel every chain pushes: Dr Lal and Metropolis both advertise it heavily. Kalpa Health's marketing team ran free home collection for a quarter and reports that bookings rose 9 percent against the previous quarter.

Marketing has asked for the budget to be doubled. The offer costs Kalpa Health a phlebotomist's visit per booking, so a doubled budget is real money, and the finance controller wants to know what it buys.

Nobody has asked who used the offer. If it mostly reached patients who already book every quarter, the 9 percent would have happened anyway and the doubled budget buys nothing.

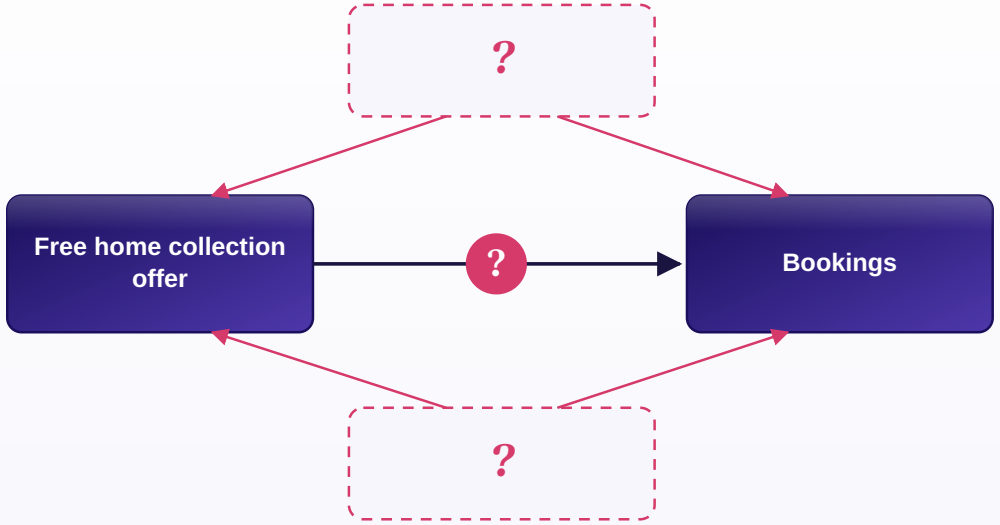
> *"Before I sign, tell me whether our offer caused our rise."*
>
> DR PRIYA MENON, COO, KALPA HEALTH

PAIN POINTS

- A lift measured by the people who want the budget
- No comparison group
- Real cost per booking

TARGET

A verdict on cause backed by a fair comparison and the lift in every segment, with a budget recommendation that states the cost of being wrong.



SEGMENT	OFFER, BEFORE	OFFER, AFTER	NO OFFER, BEFORE	NO OFFER, AFTER
?				
?				
?				

Name the two blank causes before you test anything, then fill the grid segment by segment.

IN THE INDUSTRY

In field experiments at eBay published in 2014, Blake, Nosko and Tadelis found that search ads on eBay's own brand name brought no measurable short-term benefit, and most other search spend reached frequent buyers who would have bought anyway. Who used Kalpa's free collection: patients it won, or patients who would have booked regardless?

Blake, Nosko and Tadelis, NBER Working Paper 20171 (2014). Checked 19 October 2026.

The pipeline, then four steps with less help each



1 STEP 1 · FULLY GUIDED

Reproduce marketing's 9 percent, stating which bookings, periods and cities it compares.

Done when your number matches theirs, or you can say exactly why not.

> WHY THIS, AND NOT THAT? Why reproduce their number before testing it?

2 STEP 2 · WITH A HINT

Look at who received the offer.

Hint: compare users and non-users on how often they booked before the offer began.

> WHY THIS, AND NOT THAT? Why behaviour before, and not after? What does 'before' protect you from?

3 STEP 3 · THE QUESTION ONLY

Split by city, patient type and channel. Does the lift hold in each part, and does the total still say 9?

> WHY THIS, AND NOT THAT? Why can every part fall while the total rises? One sentence, no jargon.

4 STEP 4 · OPEN

What would make you wrong? Name the comparison that would convince marketing, and what you would need to run it.

> WHY THIS, AND NOT THAT? Next quarter, who would not get the offer, and how would you choose them?

DONE LOOKS LIKE

A verdict on cause, the lift per segment, the rejected comparisons logged, and a budget recommendation with the cost of being wrong.

PUSH FURTHER

Design the next campaign so the question can be answered cleanly: who gets it, who does not, for how long, measured how.

THE VIVA WILL ASK

1. Correlation or causation: which did marketing show, and which did you?
2. One sentence a COO can carry into a board meeting.
3. Would you double the budget, and what would change your mind?

How you will be questioned

The panel, the mock and the group discussion test one thing: say what you found, how sure you are, what to do, and survive the follow-up.

THE ANSWER SHAPE

Claim
What you found, in one sentence.

↓

Evidence
The numbers, each with its denominator.

↓

Caveat
The one thing that would change the claim.

↓

Action
What Dr Menon should do, and what it costs.

“Not yet, and here is what would tell us” is a complete answer when it is true.

THE CHALLENGES LOG

WHEN	WHAT BROKE	WHAT WE TRIED	WHAT WE CHANGED, AND WHY
Phase 2	First join of invoices to payments returned almost no matches.	Printed ten ids from each file side by side.	Wrote a mapping rule; kept unmappable rows in a separate list.

Write entries as they happen. The commit history shows which kind you kept.

THE MATURITY LADDER

THIS PROGRAMME’S OWN DEVICE

4 **You recommend, and price being wrong.**
The panel tests your caveat.

3 **You say how sure, and why.**
So what should I do?

2 **You explain what moved it.**
How sure are you?

1 **You report a number.**
Compared with what?

Open at step three. Reach step four when pressed.

The presentation and the video

THE SLOT, ABOUT 25 TO 30 MINUTES

About 10 minutes
Six slides. Any member may be asked to present any slide.

About 5 minutes
Live demo from a fresh Codespace, running the notebook that produces the headline number.

10 to 15 minutes
Questions to the group, then to individuals, then to the quietest person.

THE SIX SLIDES

1. **The question**
   Dr Menon’s words and your translation.
2. **The picture**
   Your completed diagram, numbered.
3. **The claim**
   One sentence. Nothing else.
4. **The evidence**
   Two or three numbers with denominators, one chart.
5. **Caveat and alternative**
   What would change the claim; the approach you rejected.
6. **The action**
   What to do, what it costs, what being wrong costs.

THE VIDEO

One video per group. Each member walks through their own part and their own contribution. Linked from the README before the presentation slot.

GROUP	LENGTH	MINUTES EACH
Four members	About 20 minutes	5 each
Three members	20 to 30 minutes	7 to 10 each
Solo	About 20 minutes	20

If the group cannot record together, each member records a 15-minute piece on their own part, linked in order.

SPEND MINUTES ON

1. The question in your words, and what you guessed before the data.
2. One decision, the alternative you rejected, the reason. Point at the log row.
3. One number you trust, its denominator, what would break it.
4. One thing that broke. Point at the challenges entry.
5. What you would do with two more days.

NOT ON

- Reading slides aloud.
- Scrolling code. Run the cell that makes the number, move on.
- Explaining syntax.
- The install, the imports.
- Recapping the brief.
- Anything a teammate covered.

- Screen and face, name and part on the first frame, one take per section.
- Record after the freeze, from the repository as graded.
- Say numbers slowly, with denominators, so a viewer can pause and check.

07 · THE DEFENCE

The individual write-up

Ten questions, 600 to 900 words, your own words, no assistant. One file, docs/writeups.md, one section per member, each committed by its author.

1	In your own words, what was Dr Menon asking for, and what did she actually need?
2	Which part was yours, and what did you produce that the group depends on?
3	Which approaches did you consider for your part, which did you choose, why not the others?
4	Which number do you trust most, its denominator, and what would make you stop trusting it?
5	What broke, what did you try, what did you change?
6	What do you understand about a diagnostics business now that you did not before?
7	Where did the team work well, where not, what changes for Build 2?
8	Which Week 1 or 2 idea do you understand better after using it on this data?
9	Which viva question would you least like, and your answer to it?
10	With two more days, what first, and why?

WHEN

A question or two each evening. Tidy it at the end; do not start it at the end.

ONE FILE, MANY AUTHORS

A two-paragraph group summary opens the file. After that each section is yours alone. Read each other’s; do not edit them.

Rules and submission

1
YOUR GROUP

Three or four people, one brief, one public repository. Any member may be questioned alone about any number.

2
NO ASSISTANTS ON THE WORK

Not the notebooks, not the write-ups, not the video script. Appendix C says what an assistant may be used for, and every use is logged.

3
IF GENERATED WORK IS FOUND

You are questioned on every line. A line you cannot explain is a reprimand for the group, and the mark follows the handbook.

4
UNDERSTAND EVERY PART

Build it part by part. Expect “why this and not that” at any point, from anyone.

5
WHAT IS GRADED

Mini Project 1: 40 marks for the implementation and your ability to explain it, including presentation and video. Mock 30, group discussion 30, of 1000.

6
WHAT YOU HAND IN

Notebooks, data as received, the three logs, the write-ups, six slides, the video link. Appendix D is the checklist.

7
WHEN

The build freezes the evening before presentations. Video link and write-ups are in the README before your slot. Nothing merged after the freeze counts.

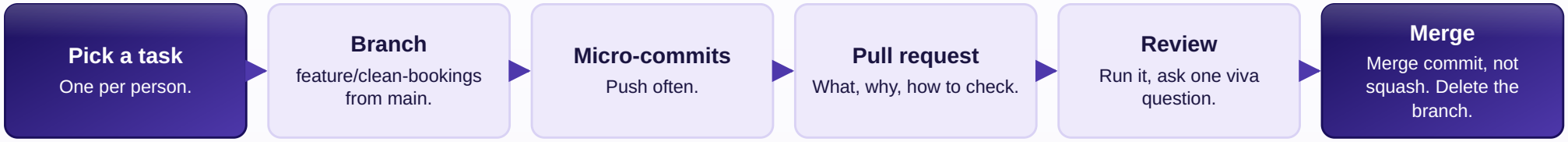
8
TOOLS

Free tools covered so far: notebooks in VS Code on Codespaces, SQL for joins, Excel for the board number, GitHub for everything.

APPENDIX A

GitHub, the way a team at work does it

One public repository per group, created before any analysis. A reviewer opens it cold and understands what you did from the README, the history and the logs.



SETTING UP, FIRST HOUR

1. **Create**
   c2-build1-brief-<letter>-group-<n>, public, collaborators added, opened in a Codespace.
2. **README first**
   Brief in one paragraph, members and parts, how to run, a Links section for the video. That is commit one.
3. **Structure**
   docs, notebooks, data, src, tests, slides in the first hour.
4. **Protect main**
   Nobody pushes to main. Everything arrives by pull request someone else approved.
5. **Data**
   data/raw as received, with a provenance README. Files over 100 MB are linked, not committed.

MICRO-COMMITS

```
Add profiling of bookings: row counts and null rates per column
```

```
Map invoice ids to payment references by stripping the INV- prefix
```

```
Reject 14 duplicate booking rows; log the rule in decisions-log.md
```

~~update~~

~~fixed stuff and added the join and changed the chart~~

~~final version 3 FINAL~~

- One change you can describe in one line. If the line needs ’and’, it is two commits.
- Commit when something runs that did not, or every thirty minutes.
- Imperative mood; say what and why.
- Clear notebook outputs before committing; one owner per notebook at a time.

REVIEWING A TEAMMATE

- Run it before you read it.
- Ask one question the panel would ask.
- Approve when you could defend it yourself.
- Never approve your own; never merge without approval.

APPENDIX B

Working as a team once the problem is defined

FOUR ROLES, ROTATED

Driver
Writes the code or slide for the current task.

Navigator
Reads every cell before it runs; predicts the output.

Scribe
Keeps the three logs current; commits them before close.

Challenger
Asks why, five times. Asks what would make the group wrong.

> *Write both approaches in the log, try the cheaper one for thirty minutes, let the result decide. A disagreement in the log earns marks; one in the corridor does not.*

THREE MODES, ONE AT A TIME

Plan
No code. Requirements, approach log, who owns what. Fifteen minutes at the start of every session.

Build
One branch, one task, one owner. New ideas go on the board, not into the driver’s ear.

Review
Fifteen minutes at the end. One thing that works, one that does not, per person. Checkpoint answers written now.

One mode at a time. Groups that build while they plan rewrite later what they wrote earlier.

THE TASK BOARD

BACKLOG	DOING	REVIEW	DONE
Written as results: ’bookings profiled with null rates’, not ’look at bookings’.	One card per person.	A pull request is open.	Merged, logged, run by someone else.

GitHub Issues with labels, or the free Projects board. The panel looks at it, so keep it true.

APPENDIX C

Testing without sinking the week, and researching well

Reconciliation

After every cleaning step: input equals clean plus rejected; revenue before equals after plus removed.

```
assert len(raw) == len(clean) +
len(rejected)
```

Joins

Row count before and after; keys matched more than once.

```
assert joined.shape[0] ==
invoices.shape[0]
```

The cold run

Once a day, restart the kernel, run everything, record the headline. If it moved, find out why before going home.

```
Kernel: Restart and Run All
```

A tenth of your time. Five pytest cases for a shared helper is plenty. Do not test pandas, charts, or anything you will delete.

RESEARCH WITHOUT AN ASSISTANT, IN THIS ORDER

1. **Your notes**
   The Week 1 decisions log and Week 2 join checks are this week's templates.
2. **Documentation**
   pandas and PostgreSQL manuals; read the examples.
3. **The industry**
   Dr Lal and Metropolis investor presentations show the tree a diagnostics board reads.
4. **Each other**
   Ask the group before the internet.
5. **The internet**
   Search the exact error in quotes; prefer dated answers; note the link in the log.

ASSISTANT: ALLOWED, AND LOGGED

- Explaining an error you already tried to read.
- Explaining a concept you then rewrite in your own words.
- Checking English in the write-up without changing meaning.

NOT ALLOWED

- Generating analysis, a notebook, a cell, a chart, a slide, a write-up, a script.
- Asking what the answer or approach should be.
- Pasting Kalpa Health data into it.

docs/ai-log.md, one line per use: date, who, what, what you did with it. Twenty honest lines is fine. An empty log next to tutorial-shaped notebooks is not.

APPENDIX D

The same five questions in businesses you walk past

The interview question this week trains is: you are in a health company now, apply what you did in retail. Name one more business per brief and the requirement that changes.

BRIEF	ANOTHER BUSINESS	THE ANALYSIS THERE	YOUR EXAMPLE
A	A gym chain: memberships up 3 against a plan of 12.	Members × visits × spend per visit, plan per branch.	
B	A food delivery app: orders fell in two zones after a payment-provider switch.	Orders per zone per week, labelled by provider; trace one order through both.	
C	A school: fee demands and receipts never match at term end.	Demanded against received by student, with an instalment timing bar.	
D	A courier: the smallest hub has the worst failed-delivery rate.	Rate with denominator, chance reference, like-with-like on route and parcel.	
E	A streaming service: a free month 'lifted sign-ups 11 percent'.	Who took it, and how often they visited the sign-up page before it existed.	

THE REPOSITORY, AS THE PANEL FINDS IT

`README.md`	Brief, members and parts, how to run, Links.
`docs/project-statement.md`	Translation note, requirements, prediction.
`docs/approach-log.md`	One row per part.
`docs/decisions-log.md`	Every cleaning decision and its effect.
`docs/challenges-log.md`	Dated entries.
`docs/writeups.md`	Group summary, one section per member.
`docs/ai-log.md`	Every assistant use.
`notebooks/01_…`	Run order, outputs cleared.
`data/raw, data/clean`	As received; what the notebooks write.
`src, tests, slides`	Shared helpers; the few tests; six slides and the Excel board number.

BEFORE YOU RECORD

- [ ] A teammate ran everything from a fresh Codespace and got the slide 3 number.
- [ ] The board matches reality.
- [ ] Every pull request approved by someone else.
- [ ] Every member has commits on all three logs.
- [ ] Write-ups: one section per member, in the author's commits.
- [ ] Approach log: two approaches in every row.
- [ ] No trainer, mark or outside student is named anywhere.